

Final Group Project Guidelines

CPE393: Introduction to Data Science with Python

Summer Semester 2026 (25 May - 7 July 2026)

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Students	Exchange students from EISEA
Group size	3 students per group
Project weight	50% of Part 2: Data Science with Python

1. Project Overview

The final project is a small end-to-end data science project completed in groups. Each group will choose a publicly available dataset, preferably from Kaggle or other sources, and use Python to define a problem, explore and clean the data, create visualizations, apply feature engineering, train at least three suitable machine learning models when applicable, evaluate the results, and discuss responsible AI practices.

Learning goals

- Apply Python programming, NumPy, Pandas, Matplotlib, Seaborn, and scikit-learn to a real dataset.
- Demonstrate the complete data science workflow from problem definition to results interpretation.
- Explain data cleaning, preprocessing, visualization, feature engineering, model training, and evaluation in clear language.
- Practice responsible use of data, machine learning, and generative AI tools.

Suitable project types

- Classification, such as predicting customer churn, disease category, sentiment class, or product rating.
- Regression, such as predicting price, score, demand, emissions, or other numerical values.
- Clustering or unsupervised analysis, such as grouping customers, countries, products, or text documents.
- EDA-focused projects are allowed only when they provide strong analysis, clear insights, and a justified reason why modeling is not suitable.

2. Dataset Requirements

- Use a publicly available dataset. Kaggle is preferred, but other public sources are allowed if they are reliable and accessible.
- The dataset must be appropriate for a medium-level data science project but should not be too trivial. Avoid toy datasets with only a few rows or no meaningful features.
- The dataset source, URL, license or usage condition, number of rows, number of columns, and target variable must be reported.
- The group must explain why the dataset is interesting and what real-world questions can be answered using it.
- Do not submit copied Kaggle notebooks or copied online solutions. The group must write and explain its own workflow.

3. Required Workflow

Required component	Expected content
Problem Definition	Define the problem clearly. State the project goal, target variable, or analysis question, and why the problem matters.
Dataset Description	Describe the dataset source, variables, size, file format, target label, if any, and important limitations.
Preprocessing and Cleaning	Handle missing values, duplicates, inconsistent formats, outliers, incorrect data types, and unnecessary columns. Explain each decision.
Exploratory Data Analysis	Use descriptive statistics and visual exploration to understand distributions, relationships, trends, and possible issues.
Data Visualization	Create clear charts using Matplotlib, Seaborn, or Plotly. Every figure must have a title, axis labels, and an interpretation.
Feature Engineering	Create, transform, encode, scale, or select features when useful. Explain why the engineered features may improve analysis or modeling.
Machine Learning Model Training	Train an appropriate model when applicable. Examples: Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, K-Means, or a simple neural network if justified.
Evaluation Metrics	Use suitable metrics. For classification: accuracy, precision, recall, F1-score, and confusion matrix. For regression: MAE, MSE/RMSE, R2. For clustering: inertia, silhouette score, or cluster interpretation.
Responsible AI Practices	Discuss data privacy, bias, fairness, limitations, data source reliability, misuse risks, and honest reporting of model performance.

4. Technical Requirements

Notebook and code

- Use Python in Google Colab, Jupyter Notebook, or other IDEs.
- Encouraged to explore the latest/new libraries (if you have knowledge)
- Use Pandas and NumPy for data loading, exploration, cleaning, and transformation.
- Use Matplotlib and/or Seaborn for visualization. Plotly may be used as an optional library.
- Use scikit-learn for machine learning model training and evaluation when modeling.
- Organize the notebook with clear markdown headings, short explanations, comments, and readable code cells.
- Make results reproducible by using a fixed `random_state` for the train-test split.
- Avoid hidden manual steps.

Recommended GitHub repository structure

CPE_Project_GroupX/

README.md

project_notebook.ipynb

project_summary.pdf

presentation_slides.pdf or presentation_slides.pptx

requirements.txt (optional but recommended)

data/ (only if the dataset license and size allow sharing)

data_link.txt (required if the full dataset is not uploaded)

figures/ (exported plots)

Other files/Information related to your project

README.md must include

- Project title and one-paragraph summary.
- Group number and member names.
- Dataset source URL and short dataset description.
- Instructions for running the notebook.
- Required libraries or environment details.
- Main results and limitations.

5. Project Summary Report

Each group must submit a concise project summary report of 15-20 pages. The report should be written clearly and should not be a direct copy of notebook outputs.

Report section	What to include
1. Project Title and Members	Project title, group number, member names, student IDs if required, and GitHub repository URL.
2. Problem Definition and Goals	Problem statement, motivation, target question, and expected outcome.
3. Dataset Description	Kaggle or public dataset source, URL, size, features, target variable, file format, license/usage condition, and dataset limitations.
4. Preprocessing	Missing values, duplicates, data type conversion, filtering, encoding, scaling, train-test split, and other cleaning decisions.
5. Data Visualization and EDA	Important charts, tables, statistics, patterns, correlations, and observations.
6. Feature Engineering	New features, transformed features, selected features, encoding, normalization, or dimensionality reduction if used.
7. Machine Learning Model Training	Algorithms used, model workflow, train-test split, hyperparameters if changed, and reason for choosing the model.
8. Evaluation Metrics and Results	Correct metrics for the task, result tables, confusion matrix or prediction plots, and honest interpretation of model strengths and weaknesses.
9. Responsible AI Practices	Bias and fairness concerns, privacy issues, limitations, source reliability, ethical risks, and how the group avoided misleading claims.
10. Conclusion and Future Work	Main findings, what the group learned, possible improvements, and next steps.
11. Team Contribution	Short table showing each member's contribution to coding, analysis, report writing, slides, and presentation.

Report formatting

- Length: 15-20 pages, excluding optional appendix.
- Use clear section headings and readable figures/tables.
- Use your own writing. Do not submit unedited AI-generated text.
- Cite the dataset source and any external references, tutorials, or code ideas used.
- Figures must be readable and must be discussed in the text.

6. Responsible AI and Generative AI Policy

Responsible AI is a required part of the project. Students may use AI tools as learning aids, but they must understand, verify, and be able to explain all submitted code, analysis, and report text.

Students must not use generative AI tools

- To submit auto-generated code or reports without understanding the logic.
- To plagiarize solutions for the final project.
- To bypass the learning process by copying and pasting answers without comprehension.
- To submit generated text without reviewing, editing, or checking correctness.

Responsible AI checklist

- Explain whether the dataset contains personal, sensitive, or potentially biased information.
- Report limitations honestly. Do not claim that a model is perfect only because one metric is high.
- Avoid data leakage. Do not use test data during preprocessing decisions or model selection.
- Use evaluation metrics that match the task and class balance.
- Mention possible real-world risks if predictions are used incorrectly.
- State whether AI tools were used for brainstorming, debugging, grammar improvement, or code explanation.

7. Deliverables and Submission

Required deliverables

Deliverable	Requirement
GitHub repository URL	Submit the repository link. The repository should contain the notebook, report, README, and project files.
Project summary report	PDF file, 15-20 pages.
Python notebook	Google Colab link or .ipynb file. The notebook must be organized and executable.
Dataset or data link	Upload dataset files only when allowed. Otherwise, provide a clear dataset download link and instructions.
Presentation slides	PDF or PPTX slides for the final presentation.
Optional supporting files	requirements.txt, exported figures, cleaned data sample, or additional scripts.

Submission deadline and naming

- Final submission deadline: 05 July 2026, 11:59 PM.
- Submit via LEB2.
- Folder/repository name: CPE_Project_GroupX, where X is the group number.
- Late submissions may receive a penalty unless prior approval is obtained.

8. Presentation and Demo

- Final project presentations will be held on 06 and 07 July 2026.
- Each group must present the project problem, dataset, workflow, visualizations, model, results, responsible AI discussion, and limitations.
- Each member must explain a meaningful part of the project and answer questions about their contribution.
- The live demo can be a short notebook walkthrough, selected code cells, GitHub repository view, or visual result explanation.
- Recommended length: 15-20 minutes per group plus Q&A. The final timing may be adjusted.

9. Evaluation Criteria

The group project is evaluated using the following 100-point rubric. The score contributes to the Group Project component of Part 2.

Category	Description	Points
Problem Definition and Goals	Clear explanation of the problem or dataset. Importance is explained, and project goals are well-defined.	10
Data Understanding and Cleaning	The dataset is carefully explored and cleaned. Missing values, duplicates, errors, and data types are handled thoughtfully.	15
Exploratory Data Analysis (EDA)	Uses statistics and visual exploration to identify patterns, relationships, trends, and issues. Insights are explained.	15
Data Visualization	Charts are effective, meaningful, labeled, readable, and interpreted correctly.	10
Modeling	Applies an appropriate ML model such as classification, regression, or clustering. Algorithm choice and workflow are explained.	15
Model Evaluation	Uses appropriate metrics and interprets results honestly,	10

Category	Description	Points
	including their limitations and potential errors.	
Code Quality and Documentation	Code is organized, readable, commented, reproducible, and supported by clear notebook markdown.	10
Presentation and Communication	Clear delivery. Team explains project flow, decisions, and outcomes, and responds well to Q&A.	10
Teamwork and Collaboration	Balanced contribution among members, smooth coordination, and evidence of collaboration.	5
Total		100